Lecture 1: Basics of Quantum Mechanics

1. Quantum State

The state of a quantum particle in one dimension is described by a complex wavefunction

$$\psi(x,t).$$

The wavefunction itself is not observable; physical information is extracted using operators.

2. Operators of Physical Observables

General principle (average measurement)

For any physical observable A represented by an operator $\hat{A}$, the average (expectation) value in the state $\psi(x,t)$ is

$$\boxed{\langle A \rangle = \int_{-\infty}^{\infty} \psi^*(x,t)\, \hat{A}\, \psi(x,t)\, dx}$$

(assuming ψ is normalized and ψ^* is the complex conjugate of the wavefunction).

Position operator

$$\boxed{\hat{x} = x}, \qquad \hat{x}\psi = x\psi.$$

Momentum operator (from a plane wave)

Consider a plane wave

$$\psi(x,t) = e^{i(kx-\omega t)}$$

where $k = \frac{p}{\hbar}$ and $\omega = \frac{E}{\hbar}$

$$\frac{\partial \psi}{\partial x} = ik\psi \quad \Rightarrow \quad -i\hbar\frac{\partial \psi}{\partial x} = \hbar k\, \psi = p\psi.$$

$$\boxed{\hat{p} = -i\hbar\frac{\partial}{\partial x}}$$

Energy operator

$$\frac{\partial \psi}{\partial t} = -i\omega\psi \quad \Rightarrow \quad i\hbar\frac{\partial \psi}{\partial t} = \hbar\omega\,\psi = E\psi.$$

$$\boxed{\hat{E} = i\hbar\frac{\partial}{\partial t}}$$

3. Schrödinger Equation

Classical energy

$$E = \frac{p^2}{2m} + V(x).$$

Replacing observables by operators:

$$\hat{E}\psi = \left(\frac{\hat{p}^2}{2m} + V\right)\psi.$$

Time-dependent Schrödinger equation

$$\boxed{i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + V(x)\psi}$$

4. Stationary Schrödinger Equation

Assume separation of variables:

$$\psi(x,t) = \psi(x)\,e^{-iEt/\hbar}.$$

Substitution gives

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)}$$

Hamiltonian operator

$$\boxed{\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)}$$

$$\hat{H}\psi(x) = E\psi(x).$$

5. Commutator

For two operators $\hat{A}$ and $\hat{B}$:

$$\boxed{[\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}}$$

For position and momentum:

$$\boxed{[\hat{x},\hat{p}] = i\hbar}$$

Commuting and non-commuting operators

Two operators $\hat{A}$ and $\hat{B}$ are said to *commute* if

$$\boxed{[\hat{A}, \hat{B}] = 0.}$$

In this case, the observables A and B are compatible: they have a common set of eigenfunctions and can be simultaneously measured with arbitrary precision.

If

$$\boxed{[\hat{A}, \hat{B}] \neq 0,}$$

the operators do not commute. The corresponding observables are incompatible, and their simultaneous measurement is limited by an uncertainty relation.

For example,

$$[\hat{x}, \hat{p}] = i\hbar \neq 0,$$

which leads to the Heisenberg uncertainty principle.

6. Heisenberg Uncertainty Principle

$$\boxed{\Delta x \, \Delta p \geq \frac{\hbar}{2}}$$

7. Probability Interpretation and Normalization

The probability density is

$$\rho(x, t) = |\psi(x, t)|^2.$$

Normalization condition:

$$\boxed{\int_{-\infty}^{\infty} |\psi(x, t)|^2 \, dx = 1}$$

8. Principle of Superposition

Since the Schrödinger equation is linear, if ψ_1 and ψ_2 are solutions, then

$$\boxed{\psi = c_1\psi_1 + c_2\psi_2}$$

is also a solution.

9. Conservation of Probability

The probability density $\rho = |\psi|^2$ satisfies the continuity equation

$$\boxed{\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0}$$

10. Probability Current Density

The probability current density is

$$\boxed{j = \frac{\hbar}{2mi}\left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x}\right)}$$

Lecture 2: Angular Momentum Operator. Mean Square Deviation (Uncertainty). Simple Quantum Systems

1 Angular Momentum Operator

The angular momentum operator is defined as

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = -i\hbar\, \mathbf{r} \times \nabla.$$

Its Cartesian components are

$$\hat{L}_x = -i\hbar \left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} \right),$$

$$\hat{L}_y = -i\hbar \left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z} \right),$$

$$\hat{L}_z = -i\hbar \left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \right).$$

2 Expectation Values and Uncertainties

2.1 Expectation Value of an Operator

For an operator $\hat{A}$, the expectation value is defined as

$$\langle A \rangle = \int \psi^*(x,t)\, \hat{A}\, \psi(x,t)\, dx.$$

2.2 Mean Square Deviation (Uncertainty)

The uncertainty (root mean square deviation) of an observable A is defined as

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2},$$

where

$$\langle A^2 \rangle = \int \psi^*(x,t)\, \hat{A}^2\, \psi(x,t)\, dx.$$

2.3 Example: Position and Momentum

For the coordinate x,

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2.$$

For the momentum p_x,

$$(\Delta p_x)^2 = \langle p_x^2 \rangle - \langle p_x \rangle^2, \qquad \hat{p}_x = -i\hbar \frac{\partial}{\partial x}.$$

These uncertainties satisfy the Heisenberg uncertainty relation

$$\boxed{\Delta x \, \Delta p_x \geq \frac{\hbar}{2}.}$$

3 Free Particle (Free Motion)

For a free particle the potential energy is zero,

$$V(x) = 0.$$

The time-independent Schrödinger equation in one dimension is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\,\psi(x).$$

This equation can be rewritten as

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0, \qquad k^2 = \frac{2mE}{\hbar^2}.$$

3.1 General Solution

The general solution of this equation is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}.$$

This solution can equivalently be written in trigonometric form:

$$\psi(x) = C\cos(kx) + D\sin(kx),$$

since

$$e^{ikx} = \cos(kx) + i\sin(kx), \qquad e^{-ikx} = \cos(kx) - i\sin(kx).$$

3.2 One-Directional Motion

The exponential terms correspond to waves propagating in opposite directions:

$$e^{ikx} \text{ — motion in the positive } x\text{-direction,}$$

$$e^{-ikx} \text{ — motion in the negative } x\text{-direction.}$$

If the particle moves only in the positive x-direction, the wave traveling in the opposite direction is absent. Setting $B = 0$, we obtain

$$\psi(x) = Ae^{ikx}.$$

3.3 Energy and Momentum

Introducing the momentum

$$p = \hbar k,$$

the energy of a free particle is

$$E = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m}.$$

4 Infinite Rectangular Potential Well

Consider a particle confined in a one-dimensional infinite rectangular potential well of width a:

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x \leq 0 \text{ or } x \geq a. \end{cases}$$

4.1 Schrödinger Equation Inside the Well

Inside the well ($0 < x < a$), the potential is zero, and the time-independent Schrödinger equation reads

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\,\psi(x).$$

This equation can be rewritten as

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0, \qquad k^2 = \frac{2mE}{\hbar^2}.$$

4.2 General Solution

The general solution inside the well is

$$\psi(x) = A\sin(kx) + B\cos(kx).$$

4.3 Boundary Conditions

Because the potential is infinite outside the well, the wave function must vanish at the boundaries:

$$\psi(0) = 0, \qquad \psi(a) = 0.$$

The first boundary condition gives

$$\psi(0) = B = 0.$$

Applying the second boundary condition,

$$\psi(a) = A\sin(ka) = 0,$$

which leads to the quantization condition

$$ka = n\pi, \qquad n = 1, 2, 3, \ldots$$

Thus,

$$k_n = \frac{n\pi}{a}.$$

4.4 Energy Spectrum

The allowed energy levels are

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2ma^2}, \qquad n = 1, 2, 3, \ldots$$

4.5 Normalized Eigenfunctions

The normalized stationary wave functions are

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \qquad 0 < x < a.}$$

Outside the well, the wave function is zero:

$$\psi_n(x) = 0, \qquad x \leq 0 \text{ or } x \geq a.$$

4.6 Normalization and Physical Interpretation

Normalization

The stationary states must satisfy the normalization condition

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 \, dx = 1.$$

Since $\psi_n(x) = 0$ outside the well, this reduces to

$$\int_0^a |\psi_n(x)|^2 \, dx = 1.$$

Assume

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{a}\right), \qquad 0 < x < a.$$

Then

$$1 = \int_0^a A^2 \sin^2\left(\frac{n\pi x}{a}\right) \, dx = A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \, dx.$$

Use the identity

$$\sin^2 u = \frac{1 - \cos(2u)}{2}.$$

Hence

$$\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \, dx = \frac{1}{2} \int_0^a \left[1 - \cos\left(\frac{2n\pi x}{a}\right)\right] dx.$$

Compute the integral:

$$\frac{1}{2}\left[x - \frac{a}{2n\pi} \sin\left(\frac{2n\pi x}{a}\right)\right]_0^a = \frac{1}{2}\left[a - \frac{a}{2n\pi} \sin(2n\pi)\right] = \frac{a}{2},$$

because $\sin(2n\pi) = 0$. Therefore,

$$1 = A^2 \frac{a}{2} \quad \Rightarrow \quad A = \sqrt{\frac{2}{a}}.$$

So the normalized eigenfunctions are

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \qquad 0 < x < a,} \qquad \psi_n(x) = 0 \text{ outside.}$$

Nodes and Quantum Number

The integer $n = 1, 2, 3, \ldots$ is the quantum number. The wave function $\psi_n(x)$ has

$$n - 1$$

nodes inside the well ($0 < x < a$), i.e. points where $\psi_n(x) = 0$.

Why the Ground-State Energy is Nonzero

The lowest allowed state corresponds to $n = 1$:

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2}.$$

It is *not* possible to have $E = 0$ inside an infinite well, because the boundary conditions $\psi(0) = \psi(a) = 0$ force the wave function to vary with x, which implies a nonzero curvature $d^2\psi/dx^2$ and thus a nonzero kinetic energy.

Equivalently, confinement to a region of size $\Delta x \sim a$ implies, by the uncertainty principle, a typical momentum uncertainty $\Delta p \sim \hbar/a$, giving an estimated kinetic energy

$$E \sim \frac{(\Delta p)^2}{2m} \sim \frac{\hbar^2}{2ma^2},$$

which is consistent with the exact result up to a numerical factor.

Lecture 3. Hermitian Operators and Evolution of Expectation Values

1 Why Hermitian Operators Are Introduced

In quantum mechanics:

- the state of a system is described by a wave function $\psi(x,t)$;
- physical observables are represented by operators.

Physical requirement: the result of any measurement must be a *real number*. This requirement leads to the concept of a *Hermitian operator*.

—

2 Transpose, Complex Conjugation, and Adjoint Operators

Let $\hat{A}$ be a linear operator.

Transpose operator. The *transpose* of $\hat{A}$, denoted by $\hat{A}^T$, is defined by

$$\int \phi\,(\hat{A}\psi)\,dV = \int (\hat{A}^T\phi)\,\psi\,dV, \qquad \forall\,\phi,\psi. \tag{1}$$

That is, $\hat{A}^T$ is obtained by interchanging the functions ϕ and ψ in the bilinear form (without complex conjugation).

Complex-conjugated operator. The *complex conjugate* of $\hat{A}$ is defined by

$$(\hat{A}\psi)^* = \hat{A}^*\,\psi^*. \tag{2}$$

Hermitian adjoint (operator with "plus"). The *Hermitian adjoint* of $\hat{A}$ (also denoted by $\hat{A}^+$) is defined as

$$\boxed{\hat{A}^\dagger \equiv (\hat{A}^T)^*} \tag{3}$$

(equivalently, $\hat{A}^\dagger = (\hat{A}^*)^T$).

With this definition, the following relation holds:

$$\int \phi^*\,(\hat{A}\psi)\,dV = \int (\hat{A}^\dagger\phi)^*\,\psi\,dV. \tag{4}$$

Hermitian operator. An operator $\hat{A}$ is *Hermitian* if

$$\hat{A}^\dagger = \hat{A}. \tag{5}$$

3 Definition of a Hermitian Operator

An operator $\hat{A}$ is called *Hermitian* if for any admissible functions ϕ and ψ:

$$\int \phi^* \left(\hat{A}\psi\right) dV = \int \left(\hat{A}\phi\right)^* \psi \, dV. \tag{6}$$

Equivalently,

$$\hat{A}^\dagger = \hat{A}. \tag{7}$$

Remark. Hermiticity is a property of the operator *together with its domain of definition.*

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4 Physical Consequences of Hermiticity

If the operator $\hat{A}$ is Hermitian, then:

- its eigenvalues are real;
- eigenfunctions corresponding to different eigenvalues are orthogonal;
- the expectation value of the observable is real:

$$\langle A \rangle \in \mathbb{R}. \tag{8}$$

—

5 Hermiticity of Basic Operators

5.1 Position Operator

$$\hat{x} = x. \tag{9}$$

Indeed,

$$\int \phi^* x \psi \, dx = \int (x\phi)^* \psi \, dx, \tag{10}$$

so the position operator is Hermitian.

—

5.2 Momentum Operator

$$\hat{p} = -i\hbar \frac{d}{dx}. \tag{11}$$

Consider:

$$\int \phi^* \hat{p} \psi \, dx = -i\hbar \int \phi^* \frac{d\psi}{dx} \, dx. \tag{12}$$

Integrating by parts:

$$= -i\hbar \left[\phi^* \psi\right]_{-\infty}^{\infty} + i\hbar \int \frac{d\phi^*}{dx} \psi \, dx. \tag{13}$$

If

$$\phi, \psi \to 0 \quad \text{at the boundaries,} \tag{14}$$

then

$$\int \phi^* \hat{p} \psi \, dx = \int (\hat{p}\phi)^* \psi \, dx. \tag{15}$$

Thus, the momentum operator is Hermitian under appropriate boundary conditions.

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5.3 Hermiticity of the Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x). \tag{16}$$

The kinetic energy operator is Hermitian, and if the potential $V(x)$ is real, the Hamiltonian $\hat{H}$ is Hermitian.

Consequences:

- the energy spectrum is real;
- eigenfunctions are orthogonal;
- the norm of the wave function is conserved.

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6 Expectation Value of an Observable

The expectation value of an observable A in the state $\psi(x,t)$ is defined as

$$\langle A \rangle = \int \psi^*(x,t) \, \hat{A} \, \psi(x,t) \, dx. \tag{17}$$

For a Hermitian operator,

$$\langle A \rangle \in \mathbb{R}. \tag{18}$$

—

7 Time Derivative of the Expectation Value

Taking the total time derivative, we obtain:

$$\frac{d}{dt}\langle A\rangle = \int \left(\frac{\partial \psi^*}{\partial t}\,\hat{A}\psi + \psi^* \hat{A}\frac{\partial \psi}{\partial t} + \psi^* \frac{\partial \hat{A}}{\partial t}\psi \right) dx. \tag{19}$$

—

7.1 Use of the Schrödinger Equation

The wave function satisfies the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi, \qquad \hat{H}^\dagger = \hat{H}. \tag{20}$$

Hence,

$$\frac{\partial \psi}{\partial t} = \frac{1}{i\hbar}\hat{H}\psi, \tag{21}$$

and for the complex conjugate:

$$\frac{\partial \psi^*}{\partial t} = -\frac{1}{i\hbar}(\hat{H}\psi)^*. \tag{22}$$

Substituting these expressions and using the Hermiticity of the Hamiltonian, we obtain:

$$\frac{d}{dt}\langle A\rangle = -\frac{1}{i\hbar}\left(\langle HA\rangle - \langle AH\rangle\right) + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle. \tag{23}$$

Finally,

$$\boxed{\frac{d}{dt}\langle A\rangle = \frac{1}{i\hbar}\langle[\hat{A},\hat{H}]\rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle} \tag{24}$$

—

8 Ehrenfest Theorem

8.1 Position Operator

For $\hat{A} = \hat{x}$:

$$[\hat{x},\hat{H}] = \frac{i\hbar}{m}\hat{p}. \tag{25}$$

Therefore,

$$\frac{d}{dt}\langle x\rangle = \frac{\langle p\rangle}{m}. \tag{26}$$

—

8.2 Momentum Operator

For $\hat{A} = \hat{p}$:

$$[\hat{p}, \hat{H}] = -i\hbar \frac{dV}{dx}. \tag{27}$$

Hence,

$$\frac{d}{dt}\langle p \rangle = -\left\langle \frac{dV}{dx} \right\rangle. \tag{28}$$

Remark. These equations represent the quantum analogue of Newton's second law.

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9 Conclusions

- Physical observables correspond to Hermitian operators;
- Hermiticity guarantees real measurement outcomes;
- The time evolution of expectation values is governed by commutators with the Hamiltonian;
- Classical mechanics emerges as a limit of quantum dynamics.

Lecture 4
Probability Current, Continuity Equation, and the Potential Step

1 Probability Conservation

The probability density is

$$\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2 = \psi^*(\mathbf{r}, t)\psi(\mathbf{r}, t). \tag{1}$$

Taking the time derivative:

$$\frac{\partial \rho}{\partial t} = \psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t}. \tag{2}$$

2 Derivation of the Continuity Equation (Landau-style)

The Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \psi + U\psi. \tag{3}$$

Complex conjugate:

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \psi^* + U\psi^*. \tag{4}$$

Multiply the first by ψ^*, the second by ψ, and subtract. The potential terms cancel:

$$i\hbar \left(\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right) = -\frac{\hbar^2}{2m} \left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* \right). \tag{5}$$

Since $\partial_t \rho = \psi^* \partial_t \psi + \psi \partial_t \psi^*$, we get

$$\frac{\partial \rho}{\partial t} = -\frac{\hbar}{2mi} \left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* \right). \tag{6}$$

Use the identity

$$\nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*) = \psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*, \tag{7}$$

to obtain

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0}, \qquad \boxed{\mathbf{j} = \frac{\hbar}{2mi} \left(\psi^* \nabla \psi - \psi \nabla \psi^* \right)}. \tag{8}$$

In one dimension:

$$j = \frac{\hbar}{2mi} \left(\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx} \right). \tag{9}$$

3 Stationary States and the Time-Independent Schrödinger Equation

Assume

$$\psi(x,t) = \phi(x)e^{-iEt/\hbar}. \tag{10}$$

Substitution into the Schrödinger equation gives

$$-\frac{\hbar^2}{2m}\frac{d^2\phi}{dx^2} + U(x)\phi = E\phi, \tag{11}$$

or equivalently

$$\frac{d^2\phi}{dx^2} + \frac{2m}{\hbar^2}\big(E - U(x)\big)\phi = 0. \tag{12}$$

4 Potential Step

$$U(x) = \begin{cases} 0, & x < 0, \\ U_0, & x > 0. \end{cases} \tag{13}$$

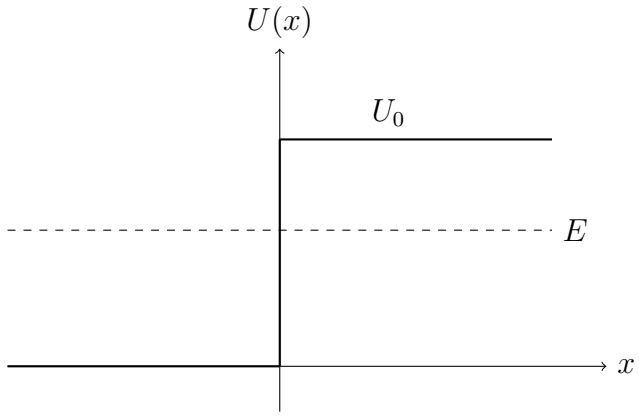


Figure 1: One-dimensional potential step.

We solve the time-independent Schrödinger equation separately in each region.

4.1 Region I: $x < 0$ (free particle)

Here $U = 0$, so

$$\phi'' + \frac{2mE}{\hbar^2}\phi = 0. \tag{14}$$

Define

$$k_1 = \frac{\sqrt{2mE}}{\hbar}, \tag{15}$$

then

$$\phi'' + k_1^2\phi = 0. \tag{16}$$

Solving the ODE explicitly

Try $\phi = e^{\lambda x}$. Substitution yields the characteristic equation:

$$\lambda^2 + k_1^2 = 0 \quad \Rightarrow \quad \lambda = \pm i k_1. \tag{17}$$

Therefore the general solution is

$$\phi_I(x) = A e^{ik_1 x} + B e^{-ik_1 x}. \tag{18}$$

For an incident wave from the left, we interpret $e^{ik_1 x}$ as right-moving (incident) and $e^{-ik_1 x}$ as left-moving (reflected). We choose the incident amplitude equal to 1:

$$\boxed{\phi_I(x) = e^{ik_1 x} + r\, e^{-ik_1 x}}. \tag{19}$$

4.2 Region II: $x > 0$ (step region)

Here $U = U_0$, so

$$\phi'' + \frac{2m(E - U_0)}{\hbar^2}\phi = 0. \tag{20}$$

4.2.1 Case A: $E > U_0$ (propagating wave)

Define

$$k_2 = \frac{\sqrt{2m(E - U_0)}}{\hbar}, \tag{21}$$

then

$$\phi'' + k_2^2 \phi = 0. \tag{22}$$

Again try $\phi = e^{\lambda x}$:

$$\lambda^2 + k_2^2 = 0 \quad \Rightarrow \quad \lambda = \pm i k_2, \tag{23}$$

so

$$\phi_{II}(x) = C e^{ik_2 x} + D e^{-ik_2 x}. \tag{24}$$

We assume there is no wave incident from $+\infty$, hence $D = 0$:

$$\boxed{\phi_{II}(x) = t\, e^{ik_2 x}}. \tag{25}$$

4.2.2 Case B: $E < U_0$ (evanescent wave, no tunneling)

Let $U_0 - E > 0$ and define

$$\kappa = \frac{\sqrt{2m(U_0 - E)}}{\hbar}. \tag{26}$$

Then

$$\phi'' - \kappa^2 \phi = 0. \tag{27}$$

Try $\phi = e^{\lambda x}$:

$$\lambda^2 - \kappa^2 = 0 \quad \Rightarrow \quad \lambda = \pm \kappa. \tag{28}$$

Thus

$$\phi_{II}(x) = C e^{\kappa x} + D e^{-\kappa x}. \tag{29}$$

To avoid divergence as $x \to +\infty$, we must set $C = 0$:

$$\boxed{\phi_{II}(x) = t\, e^{-\kappa x}}. \tag{30}$$

Remark. For $E < U_0$ in the step problem there is *no tunneling*, because the classically forbidden region extends to $+\infty$. The wave function penetrates but carries no probability flux in $x > 0$.

5 Boundary Conditions at $x = 0$

For a finite step potential, both ϕ and ϕ' are continuous:

$$\boxed{\phi_I(0) = \phi_{II}(0), \qquad \phi_I'(0) = \phi_{II}'(0)}. \tag{31}$$

5.1 Coefficients for $E > U_0$

Using $\phi_I = e^{ik_1 x} + re^{-ik_1 x}$ and $\phi_{II} = te^{ik_2 x}$:

At $x = 0$:

$$1 + r = t. \tag{32}$$

Derivatives at $x = 0$:

$$ik_1(1 - r) = ik_2 t. \tag{33}$$

Solving,

$$\boxed{r = \frac{k_1 - k_2}{k_1 + k_2}}, \qquad \boxed{t = \frac{2k_1}{k_1 + k_2}}. \tag{34}$$

5.2 Coefficients for $E < U_0$

Using $\phi_I = e^{ik_1 x} + re^{-ik_1 x}$ and $\phi_{II} = te^{-\kappa x}$:

At $x = 0$:

$$1 + r = t. \tag{35}$$

Derivatives at $x = 0$:

$$ik_1(1 - r) = -\kappa t. \tag{36}$$

Solving,

$$\boxed{r = \frac{ik_1 + \kappa}{ik_1 - \kappa}}, \qquad \boxed{t = \frac{2ik_1}{ik_1 - \kappa}}. \tag{37}$$

6 Reflection and Transmission via Probability Current

For a plane wave Ae^{ikx},

$$j = \frac{\hbar k}{m}|A|^2, \tag{38}$$

and for Ae^{-ikx} the current is negative.

6.1 Case $E > U_0$

Incoming, reflected, and transmitted currents:

$$j_{\text{in}} = \frac{\hbar k_1}{m}, \qquad j_{\text{ref}} = -\frac{\hbar k_1}{m}|r|^2, \qquad j_{\text{tr}} = \frac{\hbar k_2}{m}|t|^2. \tag{39}$$

Define

$$R = \frac{|j_{\text{ref}}|}{j_{\text{in}}} = |r|^2, \qquad T = \frac{j_{\text{tr}}}{j_{\text{in}}} = \frac{k_2}{k_1}|t|^2. \tag{40}$$

Using the expressions for r and t:

$$\boxed{R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2}, \qquad \boxed{T = \frac{4k_1k_2}{(k_1 + k_2)^2}}, \qquad \boxed{R + T = 1}. \tag{41}$$

6.2 Case $E < U_0$

In region $x > 0$ the wave function is evanescent and can be chosen real, hence

$$j_{II} = 0 \quad \Rightarrow \quad \boxed{T = 0, \quad R = 1}. \tag{42}$$

Summary

- Derived the continuity equation and defined the probability current density.
- Reduced the time-dependent Schrödinger equation to the stationary form.
- Solved the Schrödinger equation explicitly in each region of the potential step.
- Applied boundary conditions to find reflection and transmission amplitudes.
- Computed R and T using the probability current; showed $R + T = 1$ for $E > U_0$.
- Explained why for $E < U_0$ there is no tunneling for a step (infinite forbidden region).

Lecture 5: Quantum Tunneling Through a Rectangular Potential Barrier (Case $E < U_0$)

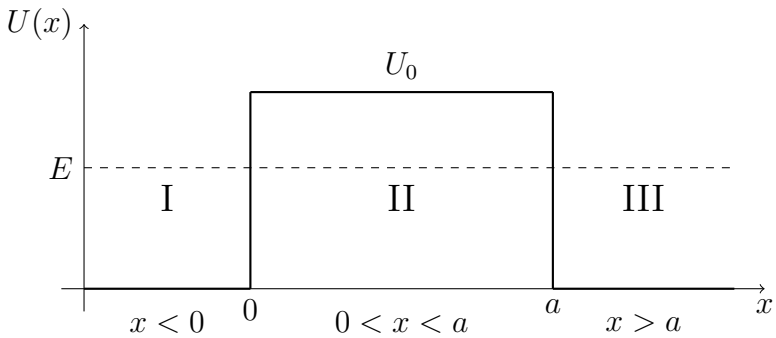
1 Potential barrier and regions

We consider a 1D rectangular barrier of height U_0 and width a:

$$U(x) = \begin{cases} 0, & x < 0 \quad \text{(Region I)} \\ U_0, & 0 < x < a \quad \text{(Region II)} \\ 0, & x > a \quad \text{(Region III)} \end{cases}$$

A particle is incident from the left with energy E. In this lecture we focus on the tunneling case:

$$E < U_0.$$



2 Time-independent Schrödinger equation

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U(x)\psi = E\psi \quad \Longleftrightarrow \quad \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\big(E - U(x)\big)\psi = 0.$$

Region-wise equations (three equations)

Define

$$k_1 = \frac{\sqrt{2mE}}{\hbar}, \qquad k_2 = \frac{\sqrt{2m(U_0 - E)}}{\hbar} \quad (E < U_0).$$

Then:

$$\begin{aligned} \textbf{Region I } (x < 0,\ U = 0): &\quad \psi_I'' + k_1^2\psi_I = 0, \\ \textbf{Region II } (0 < x < a,\ U = U_0): &\quad \psi_{II}'' - k_2^2\psi_{II} = 0, \\ \textbf{Region III } (x > a,\ U = 0): &\quad \psi_{III}'' + k_1^2\psi_{III} = 0. \end{aligned}$$

3 General solutions in Regions I–II–III

Assume incidence from the left and no incoming wave from $+\infty$:

$$\boxed{\psi_I(x) = e^{ik_1x} + A\,e^{-ik_1x}} \qquad (x<0),$$

$$\boxed{\psi_{II}(x) = B\,e^{k_2x} + B'\,e^{-k_2x}} \qquad (0<x<a),$$

$$\boxed{\psi_{III}(x) = C\,e^{ik_1x}} \qquad (x>a).$$

Unknown coefficients: A, B, B', C.

4 Boundary (matching) conditions (written as formulas)

For a *finite* potential barrier, both ψ and ψ' are continuous at each boundary.

At $x = 0$

$$\psi_I(0) = \psi_{II}(0) \implies 1 + A = B + B'. \tag{BC1}$$

$$\psi_I'(0) = \psi_{II}'(0) \implies ik_1(1-A) = k_2(B-B'). \tag{BC2}$$

At $x = a$

$$\psi_{II}(a) = \psi_{III}(a) \implies Be^{k_2a} + B'e^{-k_2a} = Ce^{ik_1a}. \tag{BC3}$$

$$\psi_{II}'(a) = \psi_{III}'(a) \implies k_2\Big(Be^{k_2a} - B'e^{-k_2a}\Big) = ik_1Ce^{ik_1a}. \tag{BC4}$$

5 Solving for C: substitution into $x = 0$ conditions and eliminating A

5.1 Step 1: Solve the $x = a$ system for B, B'

Introduce

$$X := Be^{k_2a}, \qquad Y := B'e^{-k_2a}.$$

Then (BC3)–(BC4) become

$$X + Y = Ce^{ik_1a},$$
$$k_2(X - Y) = ik_1Ce^{ik_1a}.$$

Solving:

$$X = \frac{Ce^{ik_1a}}{2}\left(1 + \frac{ik_1}{k_2}\right), \qquad Y = \frac{Ce^{ik_1a}}{2}\left(1 - \frac{ik_1}{k_2}\right).$$

Return to B, B':

$$\boxed{B = \frac{Ce^{ik_1a}}{2}\,e^{-k_2a}\left(1 + \frac{ik_1}{k_2}\right)}, \qquad \boxed{B' = \frac{Ce^{ik_1a}}{2}\,e^{k_2a}\left(1 - \frac{ik_1}{k_2}\right)}. \tag{1}$$

5.2 Step 2: Eliminate A using the $x = 0$ equations

From (BC1):

$$A = (B + B') - 1 \quad \Rightarrow \quad 1 - A = 2 - (B + B').$$

Substitute into (BC2):

$$ik_1\Big(2 - (B + B')\Big) = k_2(B - B'). \tag{2}$$

So we need $B + B'$ and $B - B'$ in terms of C.

5.3 Step 3: Compute $B + B'$ and $B - B'$ explicitly (substitution shown)

Using (1):

$$B + B' = \frac{Ce^{ik_1a}}{2}\left[e^{-k_2a}\left(1 + \frac{ik_1}{k_2}\right) + e^{k_2a}\left(1 - \frac{ik_1}{k_2}\right)\right].$$

Group terms:

$$B + B' = \frac{Ce^{ik_1a}}{2}\left[(e^{k_2a} + e^{-k_2a}) + \frac{ik_1}{k_2}(e^{-k_2a} - e^{k_2a})\right].$$

Using $\cosh z = \frac{e^z + e^{-z}}{2}$ and $\sinh z = \frac{e^z - e^{-z}}{2}$:

$$\boxed{B + B' = Ce^{ik_1a}\left[\cosh(k_2a) - i\frac{k_1}{k_2}\sinh(k_2a)\right].} \tag{3}$$

Similarly,

$$B - B' = \frac{Ce^{ik_1a}}{2}\left[e^{-k_2a}\left(1 + \frac{ik_1}{k_2}\right) - e^{k_2a}\left(1 - \frac{ik_1}{k_2}\right)\right],$$

so

$$B - B' = \frac{Ce^{ik_1a}}{2}\left[(e^{-k_2a} - e^{k_2a}) + \frac{ik_1}{k_2}(e^{-k_2a} + e^{k_2a})\right],$$

hence

$$\boxed{B - B' = Ce^{ik_1a}\left[-\sinh(k_2a) + i\frac{k_1}{k_2}\cosh(k_2a)\right].} \tag{4}$$

5.4 Step 4: Substitute into (2) and solve for C

Insert (3) and (4) into (2):

$$ik_1\left(2 - Ce^{ik_1a}\left[\cosh(k_2a) - i\frac{k_1}{k_2}\sinh(k_2a)\right]\right) = k_2\,Ce^{ik_1a}\left[-\sinh(k_2a) + i\frac{k_1}{k_2}\cosh(k_2a)\right].$$

Expand the left-hand side:

$$2ik_1 - ik_1Ce^{ik_1a}\cosh(k_2a) - \frac{k_1^2}{k_2}Ce^{ik_1a}\sinh(k_2a) = -k_2Ce^{ik_1a}\sinh(k_2a) + ik_1Ce^{ik_1a}\cosh(k_2a).$$

Move the C-terms to the right and factor Ce^{ik_1a}:

$$2ik_1 = Ce^{ik_1a}\left[2ik_1\cosh(k_2a) + \left(\frac{k_1^2}{k_2} - k_2\right)\sinh(k_2a)\right].$$

Note that $\frac{k_1^2}{k_2} - k_2 = \frac{k_1^2 - k_2^2}{k_2}$. Therefore

$$Ce^{ik_1 a} = \frac{2ik_1}{2ik_1 \cosh(k_2 a) + \frac{k_1^2 - k_2^2}{k_2} \sinh(k_2 a)}.$$

Multiplying numerator and denominator by k_2 and writing the result in the standard form gives the **transmission amplitude**:

$$\boxed{C = \frac{2k_1 k_2 \, e^{-ik_1 a}}{2k_1 k_2 \cosh(k_2 a) - i\,(k_1^2 - k_2^2) \sinh(k_2 a)}}. \tag{5}$$

6 Transmission coefficient

The 1D probability current is

$$j = \frac{\hbar}{2mi} \left(\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx} \right).$$

Since Regions I and III have the same wavenumber k_1, the transmission coefficient is

$$T = \frac{j_{\text{trans}}}{j_{\text{inc}}} = |C|^2.$$

Using (5):

$$\boxed{T = \frac{4k_1^2 k_2^2}{(k_1^2 + k_2^2)^2 \sinh^2(k_2 a) + 4k_1^2 k_2^2}}. \tag{6}$$

Equivalently,

$$\boxed{T = \frac{1}{1 + \frac{(k_1^2 + k_2^2)^2}{4k_1^2 k_2^2} \sinh^2(k_2 a)}}.$$

7 Asymptotic form (added at the end)

For an "opaque" barrier ($k_2 a \gg 1$),

$$\sinh(k_2 a) \approx \frac{1}{2} e^{k_2 a},$$

so from (6) one obtains

$$\boxed{T \approx \frac{16k_1^2 k_2^2}{(k_1^2 + k_2^2)^2} \, e^{-2k_2 a} \approx \frac{16k_1^2 k_2^2}{(k_1^2 + k_2^2)^2} \, e^{-\frac{2a\sqrt{2m(U_0 - E)}}{\hbar}} \qquad (k_2 a \gg 1).}$$

Lecture 6
Quantum Harmonic Oscillator
Power Series Termination Method

1. Problem Formulation

The harmonic oscillator potential:

$$V(x) = \frac{1}{2}m\omega^2 x^2.$$

Time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi.$$

2. Transformation to a Dimensionless Variable

Introduce the characteristic length:

$$x_0 = \sqrt{\frac{\hbar}{m\omega}}.$$

Define the dimensionless variable:

$$\xi = \frac{x}{x_0} = \sqrt{\frac{m\omega}{\hbar}}\,x.$$

Derivative Transformation

Since $x = x_0\xi$,

$$\frac{d}{dx} = \frac{1}{x_0}\frac{d}{d\xi}, \qquad \frac{d^2}{dx^2} = \frac{1}{x_0^2}\frac{d^2}{d\xi^2}.$$

Substituting into Schrödinger equation:

$$-\frac{\hbar^2}{2mx_0^2}\psi''(\xi) + \frac{1}{2}m\omega^2 x_0^2\xi^2\psi(\xi) = E\psi(\xi).$$

Multiply by $\frac{2mx_0^2}{\hbar^2}$:

$$-\psi''(\xi) + \frac{m^2\omega^2 x_0^4}{\hbar^2}\xi^2\psi(\xi) = \frac{2mx_0^2}{\hbar^2}E\psi(\xi).$$

Since $m\omega x_0^2 = \hbar$, we obtain:

$$-\psi''(\xi) + \xi^2\psi(\xi) = \lambda\psi(\xi), \qquad \lambda = \frac{2E}{\hbar\omega}.$$

Equivalent form:

$$\psi'' + (\lambda - \xi^2)\psi = 0.$$

3. Asymptotic Behavior

For $|\xi| \to \infty$:

$$\psi'' - \xi^2\psi \approx 0.$$

Assume:

$$\psi \sim e^{a\xi^2}.$$

Substitution gives:

$$4a^2 = 1, \qquad a = \pm\frac{1}{2}.$$

Normalizability requires:

$$\psi \sim e^{-\xi^2/2}.$$

4. Extracting the Gaussian Factor

Let

$$\psi(\xi) = e^{-\xi^2/2}y(\xi).$$

After differentiation:

$$y'' - 2\xi y' + (\lambda - 1)y = 0.$$

5. Power Series Solution

Assume a power series solution:

$$y(\xi) = \sum_{n=0}^{\infty} a_n\xi^n.$$

First Derivative

$$y'(\xi) = \sum_{n=1}^{\infty} na_n\xi^{n-1}.$$

Second Derivative

$$y''(\xi) = \sum_{n=2}^{\infty} n(n-1)a_n\xi^{n-2}.$$

Substitution into the Differential Equation

Substitute into

$$y'' - 2\xi y' + (\lambda - 1)y = 0 :$$

$$\sum_{n=2}^{\infty} n(n-1)a_n\xi^{n-2} - 2\xi \sum_{n=1}^{\infty} na_n\xi^{n-1} + (\lambda - 1)\sum_{n=0}^{\infty} a_n\xi^n = 0.$$

Simplify the second term:

$$2\xi \sum_{n=1}^{\infty} na_n\xi^{n-1} = 2\sum_{n=1}^{\infty} na_n\xi^n.$$

Shift the index in the first sum by letting $n \to n+2$:

$$\sum_{n=2}^{\infty} n(n-1)a_n\xi^{n-2} = \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}\xi^n.$$

Now all terms are expressed in powers of ξ^n:

$$\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}\xi^n - 2\sum_{n=1}^{\infty} na_n\xi^n + (\lambda - 1)\sum_{n=0}^{\infty} a_n\xi^n = 0.$$

Since the $n = 0$ term of the second sum vanishes, we may write:

$$2\sum_{n=1}^{\infty} na_n\xi^n = 2\sum_{n=0}^{\infty} na_n\xi^n.$$

Thus,

$$\sum_{n=0}^{\infty}\Big[(n+2)(n+1)a_{n+2} - 2na_n + (\lambda - 1)a_n\Big]\xi^n = 0.$$

Combine terms:

$$\sum_{n=0}^{\infty}\Big[(n+2)(n+1)a_{n+2} + (\lambda - 1 - 2n)a_n\Big]\xi^n = 0.$$

Since this must hold for all ξ, each coefficient must vanish:

$$(n+2)(n+1)a_{n+2} + (\lambda - 1 - 2n)a_n = 0.$$

Recurrence Relation

$$\boxed{a_{n+2} = \frac{2n+1-\lambda}{(n+2)(n+1)}\,a_n, \qquad n = 0, 1, 2, \ldots}$$

6. Termination Condition and Energy Quantization

To avoid divergence:

$$2n + 1 - \lambda = 0.$$

Thus,

$$\lambda = 2n + 1.$$

Since $\lambda = \frac{2E}{\hbar\omega}$:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right).$$

7. Hermite Polynomials

When the series terminates:

$$y(\xi) = H_n(\xi),$$

Rodrigues formula:

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n}{d\xi^n} e^{-\xi^2}.$$

Orthogonality:

$$\int_{-\infty}^{\infty} e^{-\xi^2} H_n(\xi) H_m(\xi)\, d\xi = \sqrt{\pi}\, 2^n n!\, \delta_{nm}.$$

8. Normalization

Since

$$x = x_0 \xi, \qquad dx = x_0 d\xi, \qquad x_0 = \sqrt{\frac{\hbar}{m\omega}},$$

Normalization condition:

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = 1.$$

Using

$$\psi_n(\xi) = \mathcal{N}_n e^{-\xi^2/2} H_n(\xi),$$

we obtain:

$$1 = x_0 |\mathcal{N}_n|^2 \sqrt{\pi}\, 2^n n!.$$

Therefore:

$$\mathcal{N}_n = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \frac{1}{\sqrt{2^n n!}}.$$

9. Final Normalized Wave Functions

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x \right) \exp\left(-\frac{m\omega x^2}{2\hbar} \right).$$

Lecture 7
Bra–Ket Formalism. Orbital Angular Momentum

Part I: Bra–Ket Formalism and Representations

1. Bra–Ket Notation

The state of a quantum system is represented by a vector

$$|\psi\rangle.$$

The corresponding conjugate vector is

$$\langle\psi|.$$

The scalar product of two states is

$$\langle\phi|\psi\rangle.$$

Normalization condition:

$$\langle\psi|\psi\rangle = 1.$$

2. Matrix Element of an Operator

Let $\hat{A}$ be an operator.

Its matrix element in a basis $|n\rangle$ is defined as

$$A_{mn} = \langle m|\hat{A}|n\rangle.$$

Example: Expectation Value

The expectation value of $\hat{A}$ in state $|\psi\rangle$ is

$$\langle\psi|\hat{A}|\psi\rangle.$$

In coordinate representation:

$$\langle\psi|\hat{A}|\psi\rangle = \int_{-\infty}^{\infty} \psi^*(x)(\hat{A}\psi)(x)\,dx.$$

For example,

$$\hat{x} = x,$$

$$\langle\psi|\hat{x}|\psi\rangle = \int \psi^*(x)\, x\, \psi(x)\, dx,$$

$$\hat{p} = -i\hbar\frac{d}{dx},$$

$$\langle\psi|\hat{p}|\psi\rangle = \int \psi^*(x) \left(-i\hbar\frac{d}{dx}\right) \psi(x)\, dx.$$

3. Coordinate Representation

Wave function:

$$\psi(x) = \langle x|\psi\rangle.$$

Scalar product:

$$\langle\phi|\psi\rangle = \int_{-\infty}^{\infty} \phi^*(x)\psi(x)\, dx.$$

Operators:

$$\hat{x} = x, \qquad \hat{p} = -i\hbar\frac{d}{dx}.$$

4. Momentum Representation

Wave function:

$$\phi(p) = \langle p|\psi\rangle.$$

Scalar product:

$$\langle\phi|\psi\rangle = \int_{-\infty}^{\infty} \phi^*(p)\psi(p)\, dp.$$

Operators:

$$\hat{p} = p, \qquad \hat{x} = i\hbar\frac{d}{dp}.$$

Part II: Orbital Angular Momentum

1. Definition

Classically,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}.$$

In quantum mechanics,

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}.$$

In index notation,

$$\hat{L}_i = \varepsilon_{ijk}\, \hat{x}_j\, \hat{p}_k,$$

where ε_{ijk} is the Levi–Civita symbol.

The Levi–Civita symbol is defined as:

$$\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = +1,$$

$$\varepsilon_{132} = \varepsilon_{213} = \varepsilon_{321} = -1,$$

and it is zero if any two indices are equal.

The indices (i, j, k) run over cyclic permutations of $(1, 2, 3)$, which correspond to (x, y, z).

2. Commutation Relations

$$[\hat{L}_i, \hat{L}_j] = i\hbar\varepsilon_{ijk}\hat{L}_k.$$

For example, using cyclic permutations:

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z,$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x,$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y.$$

However,

$$[\hat{L}^2, \hat{L}_z] = 0,$$

so we can choose common eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$.

3. Operator $\hat{L}_z$ in Spherical Coordinates

Using

$$x = r\sin\theta\cos\phi,$$

$$y = r\sin\theta\sin\phi,$$

$$z = r\cos\theta,$$

one obtains

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}.$$

4. Eigenfunctions of $\hat{L}_z$

$$\hat{L}_z \psi = \lambda \psi$$

gives

$$\psi(\phi) = e^{im\phi}, \qquad \lambda = \hbar m.$$

Single-valuedness:

$$e^{i2\pi m} = 1 \quad \Rightarrow \quad m \in \mathbb{Z}.$$

Thus

$$m = 0, \pm 1, \pm 2, \ldots$$

5. Total Orbital Angular Momentum

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2.$$

In spherical coordinates,

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right].$$

6. Eigenvalue Problem

$$\hat{L}^2 Y = \hbar^2 l(l+1) Y,$$
$$\hat{L}_z Y = \hbar m Y.$$

We use separation of variables:

$$Y(\theta, \phi) = \Theta(\theta)\, \Phi(\phi),$$

where

$$\Phi(\phi) = e^{im\phi}.$$

The equation for $\Theta(\theta)$ reduces to the associated Legendre equation. Regularity at $\theta = 0$ and $\theta = \pi$ implies

$$l = 0, 1, 2, \ldots$$

For each l,

$$m = -l, -l+1, \ldots, l.$$

The eigenfunctions are the spherical harmonics

$$Y_{lm}(\theta, \phi) = \Theta_{lm}(\theta)\, e^{im\phi}.$$

7. Eigenvalues

$$\hat{L}^2 \to \hbar^2 l(l+1),$$
$$\hat{L}_z \to \hbar m.$$

Lecture 8
Motion in a Central Potential

1 Central Potential

A potential is called **central** if it depends only on the distance from the origin:

$$V(\mathbf{r}) = V(r), \qquad r = \sqrt{x^2 + y^2 + z^2}.$$

Such potentials possess spherical symmetry. Examples include

- Coulomb potential
- gravitational potential
- spherically symmetric nuclear potentials

Because of this symmetry it is convenient to use **spherical coordinates**

$$(r, \theta, \phi).$$

2 Time–Independent Schrödinger Equation

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi.$$

The Laplacian in spherical coordinates is

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}.$$

3 Connection with Angular Momentum

The angular part of the Laplacian can be expressed using the square of the orbital angular momentum operator:

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right].$$

Thus the Laplacian can be written as

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}.$$

Substituting into the Schrödinger equation gives

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)-\frac{\hat{L}^2}{\hbar^2 r^2}\right]\psi+V(r)\psi=E\psi.$$

4 Separation of Variables

Due to spherical symmetry we look for solutions in the form

$$\psi(r,\theta,\phi)=R(r)Y(\theta,\phi),$$

where

- $R(r)$ is the radial part,
- $Y(\theta,\phi)$ is the angular part.

Substituting this form into the Schrödinger equation

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)-\frac{\hat{L}^2}{\hbar^2 r^2}\right]\psi+V(r)\psi=E\psi,$$

we obtain

$$-\frac{\hbar^2}{2m}\left[\frac{Y}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-\frac{R}{\hbar^2 r^2}\hat{L}^2 Y\right]+V(r)RY=ERY.$$

Dividing by RY gives

$$-\frac{\hbar^2}{2m}\left[\frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-\frac{1}{\hbar^2 r^2}\frac{\hat{L}^2 Y}{Y}\right]+V(r)=E.$$

Rearranging, we write

$$-\frac{\hbar^2}{2m}\frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+V(r)+\frac{1}{2mr^2}\frac{\hat{L}^2 Y}{Y}=E.$$

Now the first two terms depend only on r, while the last term depends only on the angular variables (θ,ϕ). Therefore the angular part must be equal to a constant. We introduce the separation constant in the form

$$\hat{L}^2 Y=\hbar^2 l(l+1)Y.$$

Thus the angular equation is

$$\hat{L}^2 Y=\hbar^2 l(l+1)Y.$$

Substituting this result back into the separated equation, we obtain

$$-\frac{\hbar^2}{2m}\frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+V(r)+\frac{\hbar^2 l(l+1)}{2mr^2}=E.$$

Multiplying by $R(r)$, we arrive at the radial equation

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-\frac{l(l+1)}{r^2}R\right]+V(r)R=ER.$$

Hence the Schrödinger equation in a central potential is separated into two independent equations:

- the angular equation
$$\hat{L}^2 Y = \hbar^2 l(l+1) Y,$$
- the radial equation
$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{l(l+1)}{r^2}R\right] + V(r)R = ER.$$

5 Angular Equation

The angular equation becomes

$$\hat{L}^2 Y = \hbar^2 l(l+1) Y.$$

Its solutions are the **spherical harmonics**

$$Y_l^m(\theta, \phi).$$

The quantum numbers take values

$$l = 0, 1, 2, 3, \ldots$$

$$m = -l, -l+1, \ldots, l.$$

In atomic physics these values are traditionally denoted by letters:

l	Orbital type
0	s
1	p
2	d
3	f
4	g
5	h

The first four letters originate from historical spectroscopic terminology:

- s – sharp
- p – principal
- d – diffuse
- f – fundamental

6 Radial Equation

The radial function satisfies

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{l(l+1)}{r^2}R\right] + V(r)R = ER.$$

7 Transformation of the Radial Equation

It is convenient to introduce a new function

$$u(r) = rR(r).$$

Then

$$R(r) = \frac{u(r)}{r}.$$

Substituting into the radial equation leads to

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left[V(r) + \frac{\hbar^2 l(l+1)}{2mr^2}\right] u = Eu.$$

8 Normalization of the Angular and Radial Functions

Since the wave function is written as

$$\psi(r,\theta,\phi) = R(r)Y_l^m(\theta,\phi),$$

the normalization condition

$$\int |\psi(r,\theta,\phi)|^2 \, d^3r = 1$$

takes the form

$$\int_0^\infty \int |R(r)|^2 |Y_l^m(\theta,\phi)|^2 \, r^2 \, dr \, d\Omega = 1,$$

where

$$d\Omega = \sin\theta \, d\theta \, d\phi.$$

It is convenient to normalize the angular and radial parts separately.
The angular functions are normalized according to

$$\int |Y_l^m(\theta,\phi)|^2 \, d\Omega = 1.$$

Therefore the radial function must satisfy

$$\int_0^\infty |R(r)|^2 r^2 \, dr = 1.$$

Using the substitution

$$u(r) = rR(r),$$

we obtain

$$R(r) = \frac{u(r)}{r},$$

and therefore the radial normalization condition becomes

$$\int_0^\infty \left|\frac{u(r)}{r}\right|^2 r^2\,dr = 1,$$

that is,

$$\int_0^\infty |u(r)|^2\,dr = 1.$$

9 Effective Potential

The quantity

$$V_{\text{eff}}(r) = V(r) + \frac{\hbar^2 l(l+1)}{2mr^2}$$

is called the **effective potential**.
The second term

$$\frac{\hbar^2 l(l+1)}{2mr^2}$$

is the **centrifugal barrier**. It arises due to the orbital angular momentum.
Thus the radial motion reduces to a one–dimensional Schrödinger equation

$$-\frac{\hbar^2}{2m}u'' + V_{\text{eff}}(r)u = Eu.$$

10 Free Particle Case

Consider the special case

$$V(r) = 0.$$

The radial equation becomes

$$u'' + \left(k^2 - \frac{l(l+1)}{r^2}\right)u = 0,$$

where

$$k^2 = \frac{2mE}{\hbar^2}.$$

This is the **spherical Bessel equation**.

11 Spherical Bessel Functions

The general solution is

$$u(r) = Aj_l(kr) + Bn_l(kr),$$

where

- j_l are spherical Bessel functions
- n_l are spherical Neumann functions

The radial function is

$$R(r) = \frac{u(r)}{r}.$$

12 Special Case $l = 0$

For

$$l = 0$$

the equation reduces to

$$u'' + k^2 u = 0.$$

The general solution is

$$u(r) = A \sin kr + B \cos kr.$$

To keep the wave function finite at $r = 0$, we set

$$B = 0.$$

Thus

$$u(r) = A \sin kr.$$

The radial function becomes

$$R(r) = \frac{A \sin kr}{r}.$$

This can also be written as

$$R(r) \propto \frac{\sin kr}{kr}.$$

Since

$$j_0(kr) = \frac{\sin kr}{kr},$$

we have

$$R(r) \propto j_0(kr).$$

13 Total Wave Function

The full wave function is

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\theta, \phi)$$

or equivalently

$$\psi(r, \theta, \phi) = \frac{u(r)}{r} Y_l^m(\theta, \phi).$$

Lecture 9
Motion in a Coulomb Potential (Hydrogen Atom)

1. Problem Setup

We consider a particle of mass m moving in a Coulomb potential:

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}$$

This potential describes the interaction between an electron and a proton in the hydrogen atom. The time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m}\Delta\psi + V(r)\psi = E\psi$$

Since the potential depends only on r, we use spherical coordinates.

Important: For bound states the energy must satisfy

$$E < 0$$

because the electron is localized near the nucleus and cannot escape to infinity.

2. Separation of Variables

We seek a solution in the form:

$$\psi(r, \theta, \phi) = R(r)Y_l^m(\theta, \phi)$$

The angular part satisfies:

$$\hat{L}^2 Y_l^m = \hbar^2 l(l+1)Y_l^m$$

Thus the angular dependence is given by spherical harmonics.

3. Radial Equation

After separation of variables, the radial Schrödinger equation can be written in terms of the reduced radial function

$$u(r) = rR(r)$$

in the form

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + V_{\text{eff}}(r)u = Eu$$

where the effective potential is

$$V_{\text{eff}}(r) = -\frac{e^2}{4\pi\varepsilon_0 r} + \frac{\hbar^2 l(l+1)}{2mr^2}$$

The first term is the Coulomb attraction, while the second term is the centrifugal barrier. For bound states, $E < 0$. It is convenient to rewrite this equation as

$$\frac{d^2u}{dr^2} + \left[\frac{2mE}{\hbar^2} + \frac{2m}{\hbar^2}\frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)}{r^2}\right] u = 0$$

and then introduce

$$k = \sqrt{-\frac{2mE}{\hbar^2}}, \qquad E < 0$$

so that the radial equation becomes

$$\frac{d^2u}{dr^2} + \left[-k^2 + \frac{2m}{\hbar^2}\frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)}{r^2}\right] u = 0$$

4. Dimensionless Variable (Detailed Derivation)

Introduce a dimensionless variable:

$$\rho = 2kr$$

Then:

$$r = \frac{\rho}{2k}$$

Compute derivatives using the chain rule:

$$\frac{d}{dr} = \frac{d\rho}{dr}\frac{d}{d\rho} = 2k\frac{d}{d\rho}$$

$$\frac{d^2}{dr^2} = (2k)^2\frac{d^2}{d\rho^2} = 4k^2\frac{d^2}{d\rho^2}$$

Substitute into the equation:

$$4k^2\frac{d^2u}{d\rho^2} + \left[-k^2 + \frac{2m}{\hbar^2}\frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)}{r^2}\right] u = 0$$

Now express everything in terms of ρ:

$$r = \frac{\rho}{2k}, \quad \frac{1}{r} = \frac{2k}{\rho}, \quad \frac{1}{r^2} = \frac{4k^2}{\rho^2}$$

Substitute:

$$4k^2u'' + \left[-k^2 + \frac{2m}{\hbar^2}\frac{e^2}{4\pi\varepsilon_0}\frac{2k}{\rho} - \frac{l(l+1)4k^2}{\rho^2}\right] u = 0$$

Divide the whole equation by $4k^2$:

$$u'' + \left[-\frac{1}{4} + \frac{me^2}{4\pi\varepsilon_0\hbar^2 k}\frac{1}{\rho} - \frac{l(l+1)}{\rho^2}\right] u = 0$$

Define:

$$\gamma = \frac{me^2}{4\pi\varepsilon_0\hbar^2 k}$$

Finally, we obtain the dimensionless equation:

$$\boxed{\frac{d^2u}{d\rho^2} + \left[-\frac{1}{4} + \frac{\gamma}{\rho} - \frac{l(l+1)}{\rho^2}\right] u = 0}$$

5. Asymptotic Behavior (Detailed Analysis)

To understand the structure of the solution, we analyze limiting cases.

(a) Behavior as $\rho \to 0$

At small ρ, the terms

$$\frac{\gamma}{\rho} \ll \frac{l(l+1)}{\rho^2}, \qquad \frac{1}{4} \ll \frac{l(l+1)}{\rho^2}$$

So the dominant term is:

$$-\frac{l(l+1)}{\rho^2}$$

Thus the equation simplifies to:

$$u'' - \frac{l(l+1)}{\rho^2} u \approx 0$$

Seek a solution of the form:

$$u \sim \rho^s$$

Then:

$$u'' = s(s-1)\rho^{s-2}$$

Substitute:

$$s(s-1)\rho^{s-2} - l(l+1)\rho^{s-2} = 0$$

$$s(s-1) - l(l+1) = 0$$

Solve:

$$s = l+1 \quad \text{or} \quad s = -l$$

Thus:

$$u \sim \rho^{l+1} \quad \text{or} \quad \rho^{-l}$$

The second solution diverges at $\rho = 0$, so we discard it.

$$\boxed{u \sim \rho^{l+1}, \quad \rho \to 0}$$

(b) Behavior as $\rho \to \infty$

At large ρ:

$$\frac{\gamma}{\rho} \to 0, \qquad \frac{l(l+1)}{\rho^2} \to 0$$

Thus the equation becomes:

$$u'' - \frac{1}{4}u \approx 0$$

Solution:

$$u \sim e^{\pm\rho/2}$$

The growing solution $e^{+\rho/2}$ is non-physical (not normalizable), so we keep:

$$\boxed{u \sim e^{-\rho/2}, \quad \rho \to \infty}$$

Conclusion

Combining both limits, the physically acceptable solution must have the form:

$$u(\rho) = \rho^{l+1} e^{-\rho/2} v(\rho)$$

where $v(\rho)$ is a regular function.

Important: This substitution removes singular and divergent behavior and leads to the Laguerre equation.

6. Solution (Laguerre Polynomials)

We set:

$$u(\rho) = \rho^{l+1} e^{-\rho/2} v(\rho)$$

Then:

$$\rho v'' + (2l + 2 - \rho)v' + (\gamma - l - 1)v = 0$$

This is the equation of associated Laguerre polynomials.

For finite (normalizable) solutions, the series must terminate:

$$\gamma - l - 1 = n_r$$

$$n = n_r + l + 1 \quad \Rightarrow \quad \gamma = n$$

$$v(\rho) = L_{n-l-1}^{2l+1}(\rho)$$

Important: The function $v(\rho)$ is exactly the associated Laguerre polynomial.

7. Energy Levels

$$\gamma = \frac{me^2}{4\pi\varepsilon_0\hbar^2 k}, \quad k = \sqrt{-\frac{2mE}{\hbar^2}}$$

From $\gamma = n$ we obtain:

$$E_n = -\frac{me^4}{2(4\pi\varepsilon_0)^2\hbar^2}\frac{1}{n^2}$$

8. Numerical Value

$$E_1 = -\frac{m_e e^4}{2(4\pi\varepsilon_0)^2\hbar^2} = -13.6\,\mathrm{eV}$$

$$E_n = -\frac{13.6}{n^2}\,\mathrm{eV}$$

9. Radial Function

The reduced radial function is obtained up to a multiplicative constant:

$$u(\rho) = C\,\rho^{l+1}e^{-\rho/2}L_{n-l-1}^{2l+1}(\rho), \qquad \rho = 2kr$$

where C is determined from normalization.
The radial wave function is

$$R_{nl}(r) = \frac{u(r)}{r}$$

10. Quantum Numbers

The solution is characterized by three quantum numbers:

- **Principal quantum number:**
$$n = 1, 2, 3, \ldots$$
- **Orbital quantum number:**
$$l = 0, 1, 2, \ldots, n-1$$
- **Magnetic quantum number:**
$$m = -l, -l+1, \ldots, l$$

They determine:

$$L^2 = \hbar^2 l(l+1), \quad L_z = \hbar m$$

11. Degeneracy

Different quantum states that have the same energy are called degenerate states.

In the hydrogen atom, the energy depends only on n, therefore all states with different l and m but the same n have the same energy.

The number of such states is:

$$\sum_{l=0}^{n-1}(2l+1) = n^2$$

Remark: This degeneracy is a special property of the Coulomb potential.

12 Energy Level Structure

$n = 1$

$$l = 0 \Rightarrow m = 0$$

States:

$$1s$$

Number of states:

$$1$$

$n = 2$

$$l = 0, 1$$

$$l = 0 \Rightarrow m = 0 \quad \Rightarrow 2s$$
$$l = 1 \Rightarrow m = -1, 0, 1 \quad \Rightarrow 2p$$

Number of states:

$$1 + 3 = 4 = 2^2$$

$n = 3$

$$l = 0, 1, 2$$

$$l = 0 \Rightarrow 3s \ (1)$$
$$l = 1 \Rightarrow 3p \ (3)$$
$$l = 2 \Rightarrow 3d \ (5)$$

Number of states:

$$1 + 3 + 5 = 9 = 3^2$$

$n = 4$

$$l = 0, 1, 2, 3$$

$$4s \ (1), \quad 4p \ (3), \quad 4d \ (5), \quad 4f \ (7)$$

Number of states:

$$1 + 3 + 5 + 7 = 16 = 4^2$$

Conclusion:

$$\sum_{l=0}^{n-1}(2l+1) = n^2$$

All these states have the same energy in the hydrogen atom.

13. Bohr Radius

Introduce the Bohr radius:

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}$$

Its numerical value is:

$$a_0 \approx 5.29 \times 10^{-11}\,\mathrm{m}$$

or

$$a_0 \approx 0.529\,\text{Å}$$

It sets the natural length scale of the hydrogen atom.

14. Relation between k and Bohr Radius

Using the energy spectrum, we obtain:

$$k = \frac{1}{na_0}$$

Thus, the Bohr radius defines the characteristic decay scale of the wave function:

$$\psi \sim e^{-r/(na_0)}$$

15. Example: Ground State (1s)

For $n = 1$, $l = 0$:

$$R_{10}(r) = 2\left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}$$

The full wave function is:

$$\psi_{100}(r,\theta,\phi) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

Probability density:

$$|\psi_{100}|^2 = \frac{1}{\pi a_0^3} e^{-2r/a_0}$$

This depends only on r, showing spherical symmetry of the ground state.

16. Summary

- Bound states correspond to $E < 0$
- The Coulomb problem reduces to a dimensionless equation
- The function $v(\rho)$ is a Laguerre polynomial
- Quantization arises from finiteness of the solution
- Energy levels: $E_n \sim -\frac{1}{n^2}$
- The Bohr radius sets the characteristic size of the atom

Lecture 10
Spin, Total Angular Momentum and Hydrogen Atom Wave Function

1. Introduction

Quantum mechanics shows that an electron possesses an intrinsic degree of freedom called **spin**.

The Stern–Gerlach experiment demonstrates that:

$$S_z = \pm\frac{\hbar}{2}$$

Physical meaning: Spin is an intrinsic angular momentum of a particle. It is not related to spatial motion but obeys the same algebra as angular momentum.

2. Spin Functions

We introduce spin functions:

$$\chi_{sm_s}$$

They satisfy:

$$\hat{S}^2\chi_{sm_s} = \hbar^2 s(s+1)\chi_{sm_s}$$
$$\hat{S}_z\chi_{sm_s} = \hbar m_s\chi_{sm_s}$$

Here:

- s is the spin quantum number, which determines the magnitude of the spin,
- m_s is the magnetic spin quantum number, i.e. the projection of the spin on the z-axis.

Allowed values of m_s are:

$$m_s = -s, -s+1, \ldots, s$$

Thus, for a given s, the spin projection can take $2s+1$ possible values.

3. Electron Spin

For the electron:

$$s = \frac{1}{2}, \quad m_s = \pm\frac{1}{2}$$

Basis spinors:

$$\chi_{1/2,+1/2} = \begin{pmatrix}1\\0\end{pmatrix}, \quad \chi_{1/2,-1/2} = \begin{pmatrix}0\\1\end{pmatrix}$$

General state:

$$\chi = \begin{pmatrix}a\\b\end{pmatrix}, \quad |a|^2 + |b|^2 = 1$$

4. Spin Operators and Pauli Matrices

Commutation relations:

$$[\hat{S}_i, \hat{S}_j] = i\hbar\varepsilon_{ijk}\hat{S}_k$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical meaning: The Pauli matrices provide a matrix representation of spin operators and describe how spin states transform under measurements along different directions. Relation:

$$\hat{S}_i = \frac{\hbar}{2}\sigma_i$$

Example:

$$\sigma_z\chi_{1/2,+1/2} = \chi_{1/2,+1/2}, \quad \sigma_z\chi_{1/2,-1/2} = -\chi_{1/2,-1/2}$$

Action of σ_z on the spin functions:

$$\sigma_z\chi_{1/2,+1/2} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \chi_{1/2,+1/2}$$

$$\sigma_z\chi_{1/2,-1/2} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -\chi_{1/2,-1/2}$$

Final result:

$$\sigma_z\chi_{1/2,\pm1/2} = \pm\chi_{1/2,\pm1/2}$$

5. Orbital Angular Momentum

$$\hat{L}^2 = \hbar^2 l(l+1), \quad m = -l, \ldots, l$$

6. Total Angular Momentum

$$\vec{J} = \vec{L} + \vec{S}$$

Physical meaning: The total angular momentum describes the complete rotational motion of the electron, including both its orbital motion around the nucleus and its intrinsic spin. Commutation relations:

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$$

Eigenvalues:

$$\hat{J}^2|j, m_j\rangle = \hbar^2 j(j+1)|j, m_j\rangle$$

$$\hat{J}_z|j, m_j\rangle = \hbar m_j|j, m_j\rangle$$

$$j = l \pm \frac{1}{2}, \quad m_j = -j, \ldots, j$$

7. Wave Function with Spin

Without spin:

$$\psi_{nlm}(r, \theta, \varphi) = R_{nl}(r)Y_l^m(\theta, \varphi)$$

With spin:

$$\Psi(\mathbf{r}) = \psi_{nlm}(\mathbf{r})\chi_{sm_s}$$

8. Basis States

Uncoupled basis:

$$|l, m\rangle \, |s, m_s\rangle$$

Identification with wavefunctions:

$$|l, m\rangle = Y_l^m(\theta, \phi), \qquad |s, m_s\rangle = \chi_{sm_s}$$

Thus, the total wavefunction is written as:

$$\Psi(\mathbf{r}, s) = R_{nl}(r) \, Y_l^m(\theta, \phi) \, \chi_{sm_s}$$

9. Coupled Basis

The coupled basis is defined by the total angular momentum:

$$|j, m_j\rangle$$

These states are eigenstates of the operators $\hat{J}^2$ and $\hat{J}_z$:

$$\hat{J}^2|j, m_j\rangle = \hbar^2 j(j+1) \, |j, m_j\rangle, \qquad \hat{J}_z|j, m_j\rangle = \hbar m_j \, |j, m_j\rangle$$

In the coordinate representation, the coupled states correspond to linear combinations of orbital and spin parts:

$$\Psi_{jm_j}(\mathbf{r}, s) = R_{nl}(r) \sum_{m, m_s} C_{lm\,sm_s}^{jm_j} \, Y_l^m(\theta, \phi) \, \chi_{sm_s}$$

In the spin-angular representation:

$$|j, m_j\rangle = \sum_{m, m_s} C_{lm\,sm_s}^{jm_j} \, |l, m\rangle \, |sm_s\rangle$$

Thus, unlike the uncoupled basis, the coupled states are superpositions of $Y_l^m(\theta, \phi) \, \chi_{sm_s}$.

10. Clebsch–Gordan Coefficients

The addition of orbital and spin angular momenta is given by:

$$|j, m_j\rangle = \sum_{m, m_s} C_{lm\,sm_s}^{jm_j} \, |l, m\rangle |s, m_s\rangle$$

Equivalently, Clebsch–Gordan coefficients can be written as scalar products:

$$C_{lm\,sm_s}^{jm_j} = \langle l, m; \, s, m_s \, | \, j, m_j\rangle$$

Thus, the coefficients represent the projection of the coupled state onto the uncoupled basis:

$$|j, m_j\rangle \;\leftrightarrow\; |l, m\rangle |s, m_s\rangle$$

Selection rule:

$$m_j = m + m_s$$

Only terms satisfying this condition contribute to the sum.

11. Hydrogen Atom Wave Function

$$\Psi_{nljm_j}(r,\theta,\varphi) = R_{nl}(r) \sum_{m,m_s} C^{jm_j}_{lm\,sm_s} Y_l^m(\theta,\varphi)\chi_{sm_s}$$

12. Special Cases

1s state:

$$\Psi = R_{10}(r) Y_0^0 \chi_{sm_s}$$

Since $l = 0$, the total angular momentum is determined only by spin:

$$j = \tfrac{1}{2}, \quad m_j = m_s$$

The Clebsch–Gordan coefficient is equal to 1:

$$\langle 0, 0;\, \tfrac{1}{2}, m_s \,|\, \tfrac{1}{2}, m_s \rangle = 1$$

Thus, no mixing occurs.
2p state: states are mixtures via Clebsch–Gordan coefficients.

13. Spin–Orbit Interaction

$$\hat{H}_{SO} \propto \vec{L} \cdot \vec{S}$$

$$\vec{L} \cdot \vec{S} = \frac{1}{2}(J^2 - L^2 - S^2)$$

This leads to fine structure splitting of energy levels.

14. Summary

$$\Psi = R_{nl}(r) \sum C\, Y_l^m \chi_{sm_s}$$

The wave function includes spatial motion, spin, and coupling between angular momenta.

Lecture 11
Non-Degenerate Perturbation Theory

1. Problem Formulation

Consider the stationary Schrödinger equation:

$$\hat{H}\psi_n = E_n\psi_n$$

Assume the Hamiltonian has the form:

$$\hat{H} = \hat{H}_0 + \lambda V, \quad \lambda \ll 1$$

where $\hat{H}_0$ is solvable and V is a small perturbation.
The spectrum is assumed to be non-degenerate:

$$E_n^{(0)} \neq E_m^{(0)}, \quad n \neq m$$

Thus, each energy level corresponds to a single eigenfunction.

2. Unperturbed Problem

$$\hat{H}_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

The eigenfunctions satisfy:

$$\int \psi_m^{(0)*}\psi_n^{(0)}\, d\tau = \delta_{mn}$$

They form a complete orthonormal set and can be used as a basis.

3. Expansions

We seek the solution in the form of series:

$$\psi_n = \psi_n^{(0)} + \lambda\psi_n^{(1)} + \lambda^2\psi_n^{(2)} + \cdots$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots$$

The corrections are assumed to be small.

4. Substitution

$$(\hat{H}_0 + \lambda V)\psi_n = E_n\psi_n$$

Substitute the expansions for the wave function and energy:

$$(\hat{H}_0 + \lambda V)(\psi_n^{(0)} + \lambda\psi_n^{(1)} + \cdots) = (E_n^{(0)} + \lambda E_n^{(1)} + \cdots)(\psi_n^{(0)} + \lambda\psi_n^{(1)} + \cdots)$$

Collecting terms of the same order in λ, we obtain equations for each order.
0th-order terms (without λ):

$$\hat{H}_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

1st-order terms with λ (with λ^1):

$$\hat{H}_0\psi_n^{(1)} + V\psi_n^{(0)} = E_n^{(0)}\psi_n^{(1)} + E_n^{(1)}\psi_n^{(0)}$$

$$(\hat{H}_0 - E_n^{(0)})\psi_n^{(1)} = E_n^{(1)}\psi_n^{(0)} - V\psi_n^{(0)}$$

This is an inhomogeneous equation for $\psi_n^{(1)}$.

5. First-Order Energy Correction

Multiply the first-order equation by $\psi_n^{(0)*}$ and integrate:

$$\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau - E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)} d\tau = E_n^{(1)} \int \psi_n^{(0)*} \psi_n^{(0)} d\tau - \int \psi_n^{(0)*} V \psi_n^{(0)} d\tau$$

Now we use the Hermiticity of $\hat{H}_0$:

$$\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau = \int (\hat{H}_0 \psi_n^{(0)})^* \psi_n^{(1)} d\tau$$

Since $\psi_n^{(0)}$ is an eigenfunction of $\hat{H}_0$:

$$\hat{H}_0 \psi_n^{(0)} = E_n^{(0)} \psi_n^{(0)}$$

we obtain:

$$\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau = E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)} d\tau$$

Thus, the first two terms cancel:

$$\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau - E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)} d\tau = 0$$

and we finally obtain:

$$E_n^{(1)} = \int \psi_n^{(0)*} \, V \, \psi_n^{(0)} \, d\tau$$

6. Normalization Condition

$$\int |\psi_n|^2 d\tau = 1$$

This leads to:

$$\int \psi_n^{(0)*} \psi_n^{(1)} d\tau = 0$$

The correction is orthogonal to the unperturbed state.

7. Expansion of the Correction

$$\psi_n^{(1)} = \sum_{m \neq n} c_m \psi_m^{(0)}$$

The term with $m = n$ is absent due to the orthogonality condition.

8. Determination of Coefficients

Consider again the first-order equation:

$$(\hat{H}_0 - E_n^{(0)}) \psi_n^{(1)} = E_n^{(1)} \psi_n^{(0)} - V \psi_n^{(0)}$$

Substitute the expansion:

$$\psi_n^{(1)} = \sum_{m \neq n} c_m \psi_m^{(0)}$$

Multiply by $\psi_m^{(0)*}$ and integrate:

$$\int \psi_m^{(0)*} (\hat{H}_0 - E_n^{(0)}) \psi_n^{(1)} \, d\tau = \int \psi_m^{(0)*} \left(E_n^{(1)} - V \right) \psi_n^{(0)} \, d\tau$$

Using:

$$\hat{H}_0 \psi_m^{(0)} = E_m^{(0)} \psi_m^{(0)}$$

we obtain:

$$(E_m^{(0)} - E_n^{(0)})c_m = -\int \psi_m^{(0)*} V \psi_n^{(0)} d\tau$$

Thus:

$$c_m = \frac{\int \psi_m^{(0)*} V \psi_n^{(0)} d\tau}{E_n^{(0)} - E_m^{(0)}}$$

The coefficients are determined by matrix elements of the perturbation.

9. First-Order Wave Function

$$\psi_n^{(1)} = \sum_{m \neq n} \frac{\int \psi_m^{(0)*} \, V \, \psi_n^{(0)} \, d\tau}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}$$

The correction is expressed as a superposition of unperturbed eigenfunctions.

10. Second-Order Energy Correction

$$\hat{H}_0 \psi_n^{(2)} + V \psi_n^{(1)} = E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}$$

Multiplying by $\psi_n^{(0)*}$ and integrating, one obtains:

$$E_n^{(2)} = \int \psi_n^{(0)*} \, V \, \psi_n^{(1)} \, d\tau$$

11. Final Expression

$$E_n^{(2)} = \sum_{m \neq n} \frac{\left| \int \psi_m^{(0)*} \, V \, \psi_n^{(0)} \, d\tau \right|^2}{E_n^{(0)} - E_m^{(0)}}$$

The second-order correction arises due to mixing with other states.

Lecture 12
Hydrogen-like Systems and Helium Atom

1. Hydrogen-like Systems

A hydrogen-like system consists of a nucleus with charge $+Ze$ and a single electron. Typical examples are

$$H, \qquad He^+, \qquad Li^{2+}, \qquad Be^{3+}.$$

Since only one electron is present, the motion is described by a single-particle Schrödinger equation in the Coulomb field of the nucleus.

The Coulomb potential is

$$V(r) = -\frac{Ze^2}{4\pi\varepsilon_0 r}.$$

The Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta - \frac{Ze^2}{4\pi\varepsilon_0 r}.$$

The stationary Schrödinger equation has the form

$$-\frac{\hbar^2}{2m}\Delta\psi - \frac{Ze^2}{4\pi\varepsilon_0 r}\psi = E\psi.$$

Because the potential depends only on the distance r, the problem has spherical symmetry. Therefore the wave function can be written as

$$\psi(r,\theta,\varphi) = R_{nl}(r)Y_{lm}(\theta,\varphi),$$

where $R_{nl}(r)$ is the radial part and $Y_{lm}(\theta,\varphi)$ are spherical harmonics.

After solving the radial equation (in the same way as for the hydrogen atom, with the inclusion of the nuclear charge Z), one obtains the well-known energy spectrum

$$E_n = -\frac{me^4Z^2}{2(4\pi\varepsilon_0)^2\hbar^2 n^2} = -13.6\,\text{eV}\,\frac{Z^2}{n^2}.$$

Thus the energy is proportional to Z^2, which means that the electron is more strongly bound for larger nuclear charge.

For example,

$$E_1(H) = -13.6\,\text{eV}, \qquad E_1(He^+) = -54.4\,\text{eV}, \qquad E_1(Li^{2+}) = -122.4\,\text{eV}.$$

Hence hydrogen-like systems are exactly solvable and play a fundamental role in atomic physics.

2. Helium Atom

The helium atom contains two electrons. Its Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2m}(\Delta_1 + \Delta_2) - \frac{2e^2}{4\pi\varepsilon_0 r_1} - \frac{2e^2}{4\pi\varepsilon_0 r_2} + \frac{e^2}{4\pi\varepsilon_0 r_{12}},$$

where

$$r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|.$$

The first term represents the kinetic energy of the two electrons. The next two terms describe the attraction of each electron to the nucleus. The last term is the Coulomb repulsion between the electrons.

The main difficulty comes from the interaction term

$$\frac{e^2}{4\pi\varepsilon_0 r_{12}},$$

because it couples the coordinates of the two electrons. For this reason, exact separation of variables is impossible, unlike in the hydrogen atom.

3. Zero Approximation and Separation

To construct an approximate solution, we split the Hamiltonian into two parts:

$$\hat{H} = \hat{H}_0 + \hat{V},$$

where

$$\hat{H}_0 = \hat{H}_1 + \hat{H}_2, \qquad \hat{V} = \frac{e^2}{4\pi\varepsilon_0 r_{12}}.$$

Here

$$\hat{H}_1 = -\frac{\hbar^2}{2m}\Delta_1 - \frac{2e^2}{4\pi\varepsilon_0 r_1}, \qquad \hat{H}_2 = -\frac{\hbar^2}{2m}\Delta_2 - \frac{2e^2}{4\pi\varepsilon_0 r_2}.$$

In the zero approximation we neglect the electron-electron repulsion and solve

$$(\hat{H}_1 + \hat{H}_2)\psi(1,2) = E\psi(1,2).$$

We look for the solution in the product form

$$\psi(1,2) = \psi_a(1)\psi_b(2),$$

which means that electron 1 is in the one-particle state a, and electron 2 is in the one-particle state b.

Substituting this form into the equation, we obtain

$$\psi_b(2)\hat{H}_1\psi_a(1) + \psi_a(1)\hat{H}_2\psi_b(2) = E\psi_a(1)\psi_b(2).$$

Dividing by $\psi_a(1)\psi_b(2)$ gives

$$\frac{\hat{H}_1\psi_a(1)}{\psi_a(1)} + \frac{\hat{H}_2\psi_b(2)}{\psi_b(2)} = E.$$

The first term depends only on the coordinates of electron 1, while the second depends only on the coordinates of electron 2. Therefore each term must be equal to a constant:

$$\hat{H}_1\psi_a = E_a\psi_a, \qquad \hat{H}_2\psi_b = E_b\psi_b,$$

and

$$E = E_a + E_b.$$

Thus, in the zero approximation, the helium atom is reduced to two independent hydrogen-like problems with $Z = 2$.

For the ground state, both electrons occupy the $1s$ state:

$$\psi^{(0)}(1,2) = \psi_{1s}(1)\psi_{1s}(2).$$

The energy of one electron is

$$E_{1s} = -54.4\,\text{eV},$$

therefore

$$E^{(0)} = 2E_{1s} = -108.8\,\text{eV}.$$

This value is too low because the repulsion between electrons has not yet been taken into account.

4. Perturbation Theory

Now the interaction between electrons is treated as a perturbation:

$$\hat{V} = \frac{e^2}{4\pi\varepsilon_0 r_{12}}.$$

The first-order correction to the energy is

$$E^{(1)} = \int \psi^{(0)*}(1,2)\frac{e^2}{4\pi\varepsilon_0 r_{12}}\psi^{(0)}(1,2)\,d\tau_1 d\tau_2.$$

After evaluation, one obtains

$$E^{(1)} \approx 34\,\text{eV}.$$

Hence the total energy in first-order perturbation theory is

$$E \approx -108.8 + 34 = -74.8\,\text{eV}.$$

The experimental value is approximately

$$E_{\text{exp}} \approx -79\,\text{eV}.$$

Thus even the first approximation gives a reasonable result, although it does not fully include electron correlation.

5. Antisymmetry of the Wave Function

Electrons are identical particles and, at the same time, they are fermions. For identical fermions, quantum mechanics requires that the total wave function must be antisymmetric under particle exchange:

$$\Psi(1,2) = -\Psi(2,1).$$

This means that if we interchange electron 1 and electron 2, the wave function changes sign. The physical state remains the same, but the mathematical form of the wave function acquires a minus sign.

This antisymmetry requirement is one of the basic properties of multi-electron systems.

6. Pauli Exclusion Principle

An important consequence of antisymmetry is the Pauli exclusion principle.

Pauli exclusion principle: two electrons cannot occupy the same quantum state.

Indeed, if two electrons are placed in the same one-particle state, then the antisymmetric combination gives

$$\Psi = 0.$$

This does not mean that the zero wave function is an allowed physical state. It means that the probability of finding the system in such a configuration is zero. Therefore such a state is forbidden.

7. Slater Determinant

To construct an antisymmetric wave function automatically, one uses the Slater determinant.

Two-electron case

For two electrons,

$$\Psi(1,2) = \frac{1}{\sqrt{2!}} \begin{vmatrix} \phi_a(1) & \phi_b(1) \\ \phi_a(2) & \phi_b(2) \end{vmatrix}.$$

Expanding the determinant, we obtain

$$\Psi(1,2) = \frac{1}{\sqrt{2}} \big[\phi_a(1)\phi_b(2) - \phi_a(2)\phi_b(1)\big].$$

Here:

- ϕ_a and ϕ_b are single-particle states (orbitals),
- the labels a, b indicate which orbital is taken,
- the arguments $(1), (2)$ indicate the electron number.

Thus $\phi_a(1)$ means: the value of the one-particle state a for electron 1.

Each one-particle state may contain both spatial and spin dependence.

Why the determinant gives antisymmetry

If we interchange electrons 1 and 2, then we interchange two rows of the determinant. From linear algebra, a determinant changes sign under the interchange of two rows. Therefore,

$$\Psi(2,1) = -\Psi(1,2).$$

Hence the Slater determinant automatically satisfies the antisymmetry requirement.

Connection with the Pauli principle

If $\phi_a = \phi_b$, then two columns of the determinant are identical. A determinant with two identical columns is zero, so

$$\Psi = 0.$$

This is exactly how the Pauli exclusion principle is implemented mathematically.

8. General Case: N Electrons

For N electrons, the Slater determinant is written as

$$\Psi(1,\ldots,N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(1) & \phi_2(1) & \ldots & \phi_N(1) \\ \phi_1(2) & \phi_2(2) & \ldots & \phi_N(2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(N) & \phi_2(N) & \ldots & \phi_N(N) \end{vmatrix}.$$

Here:

- $\phi_1, \phi_2, \ldots, \phi_N$ are the chosen one-particle orbitals,
- the index of ϕ_k labels the orbital,
- the argument (i) labels the electron.

So the matrix element

$$\phi_k(i)$$

means the value of the k-th orbital for the i-th electron.

In this determinant:

- each row corresponds to one electron,
- each column corresponds to one orbital.

When two electrons are exchanged, two rows are exchanged, so the determinant changes sign. When two orbitals are the same, two columns are identical, so the determinant vanishes.

9. Total Wave Function

For a two-electron system, the total wave function can be written as

$$\Psi = \Psi_{\text{space}}\chi_{\text{spin}}.$$

Since the total wave function must be antisymmetric, the symmetry of the spatial part and the symmetry of the spin part must be opposite:

- if the spin part is antisymmetric, the spatial part must be symmetric;
- if the spin part is symmetric, the spatial part must be antisymmetric.

This is the key idea in the discussion of parahelium and orthohelium.

10. Parahelium and Orthohelium

Parahelium ($S = 0$)

Parahelium corresponds to the singlet state with total spin

$$S = 0.$$

Its spin function is antisymmetric:

$$\chi_0 = \frac{1}{\sqrt{2}}\big[\chi_+(1)\chi_-(2) - \chi_+(2)\chi_-(1)\big].$$

Therefore the spatial part must be symmetric:

$$\Psi_{\text{space}}(1,2) = \Psi_{\text{space}}(2,1).$$

This case corresponds to the ground state of helium, where both electrons occupy the $1s$ state.

Orthohelium ($S = 1$)

Orthohelium corresponds to the triplet state with total spin

$$S = 1.$$

Its spin functions are symmetric:

$$\chi_1^{(1)} = \chi_+(1)\chi_+(2),$$

$$\chi_1^{(0)} = \frac{1}{\sqrt{2}}\big[\chi_+(1)\chi_-(2) + \chi_+(2)\chi_-(1)\big],$$

$$\chi_1^{(-1)} = \chi_-(1)\chi_-(2).$$

Therefore the spatial part must be antisymmetric:

$$\Psi_{\text{space}}(1,2) = -\Psi_{\text{space}}(2,1).$$

Thus parahelium and orthohelium differ by the symmetry of their spin and spatial parts, which leads to different physical properties.

Lecture 13
Quantum Scattering Theory (Core Concepts)

1. Schrödinger Equation

We consider scattering states with positive energy:

$$E > 0$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

At large distances ($V(r) \to 0$):

$$\nabla^2\psi + k^2\psi = 0, \quad k = \frac{\sqrt{2mE}}{\hbar}$$

2. Asymptotic Wave Function

$$\psi(\mathbf{r}) \sim e^{ikz} + f(\theta)\frac{e^{ikr}}{r}$$

- e^{ikz} — incident plane wave
- $\frac{e^{ikr}}{r}$ — scattered spherical wave
- $f(\theta)$ — scattering amplitude

Interpretation: all scattering information is contained in $f(\theta)$.

3. Probability Current

$$\mathbf{j} = \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*)$$

$$j_{\text{inc}} = \frac{\hbar k}{m}, \qquad j_{\text{sc}} = \frac{\hbar k}{m}\frac{|f(\theta)|^2}{r^2}$$

Interpretation: the current describes the flow of particles.

4. Differential Cross Section

$$\frac{d\sigma}{d\Omega} = \frac{\text{scattered flux}}{\text{incident flux}}$$

$$\boxed{\frac{d\sigma}{d\Omega} = |f(\theta)|^2}$$

Meaning: cross section per unit solid angle
Units: m^2/sr

$$1\,\text{barn} = 1\,\text{bn} = 10^{-28}\,\mathrm{m}^2, \qquad 1\,\text{mbn} = 10^{-3}\,\text{bn}$$

(sr is dimensionless; bn and mbn are abbreviations used for convenience)

5. Total Elastic Cross Section (Integral over all angles)

$$\sigma = \int |f(\theta)|^2 d\Omega$$

Meaning: total scattering probability (effective interaction area)

6. Central Potential

$$V = V(r)$$

The system is spherically symmetric; therefore angular momentum is conserved.

7. Partial Wave Expansion

$$\psi(r, \theta) = \sum_{l=0}^{\infty} (2l + 1) R_l(r) P_l(\cos\theta)$$

Interpretation: decomposition into angular momentum channels.

8. Legendre Polynomials

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l, \quad x = \cos\theta$$

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{1}{2}(3x^2 - 1)$$

$$\int_{-1}^{1} P_l(x) P_{l'}(x)\, dx = \frac{2}{2l + 1} \delta_{ll'}$$

9. Associated Legendre Polynomials and Spherical Harmonics

$$P_l^m(x) = (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_l(x)$$

$$Y_l^m(\theta, \phi) = (-1)^m \sqrt{\frac{2l + 1}{4\pi} \frac{(l - m)!}{(l + m)!}} P_l^m(\cos\theta) e^{im\phi}$$

$$Y_l^0(\theta, \phi) = \sqrt{\frac{2l + 1}{4\pi}} P_l(\cos\theta)$$

Only $m = 0$ contributes due to axial symmetry.

10. Phase Shifts

At large distances:

$$R_l(r) \sim \frac{1}{kr} \sin\left(kr - \frac{l\pi}{2} + \delta_l\right)$$

Interpretation: δ_l is the phase shift caused by the potential.

11. Relation Between Potential and Phase Shifts

$$\frac{d^2u_l}{dr^2} + \left(k^2 - \frac{l(l+1)}{r^2} - \frac{2m}{\hbar^2}V(r)\right)u_l = 0, \qquad u_l = rR_l$$

Outside the interaction region ($V = 0$):

$$\frac{d^2u_l}{dr^2} + \left(k^2 - \frac{l(l+1)}{r^2}\right)u_l = 0$$

General solution:

$$u_l(r) \propto \cos\delta_l \, j_l(kr) - \sin\delta_l \, n_l(kr)$$

Asymptotically:

$$u_l(r) \sim \sin\left(kr - \frac{l\pi}{2} + \delta_l\right)$$

Key idea: the potential shifts the phase of the wave.

12. Scattering Amplitude

$$f(\theta) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)e^{i\delta_l}\sin\delta_l \, P_l(\cos\theta)$$

13. Total Cross Section

$$\sigma = \frac{4\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)\sin^2\delta_l$$

14. Low-Energy Limit

$$l = 0$$

$$\sigma \approx \frac{4\pi}{k^2}\sin^2\delta_0$$

Interpretation: scattering becomes isotropic.

Conclusion

Scattering is determined by phase shifts:

$$\delta_l$$

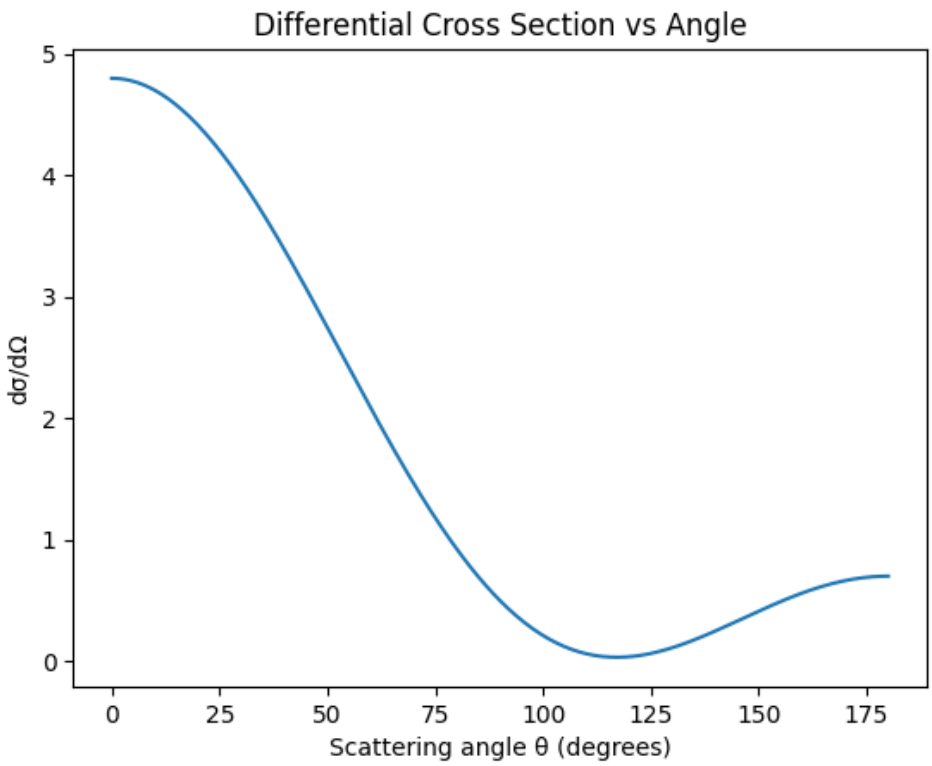


Figure 1: Differential cross section as a function of scattering angle θ (in degrees).

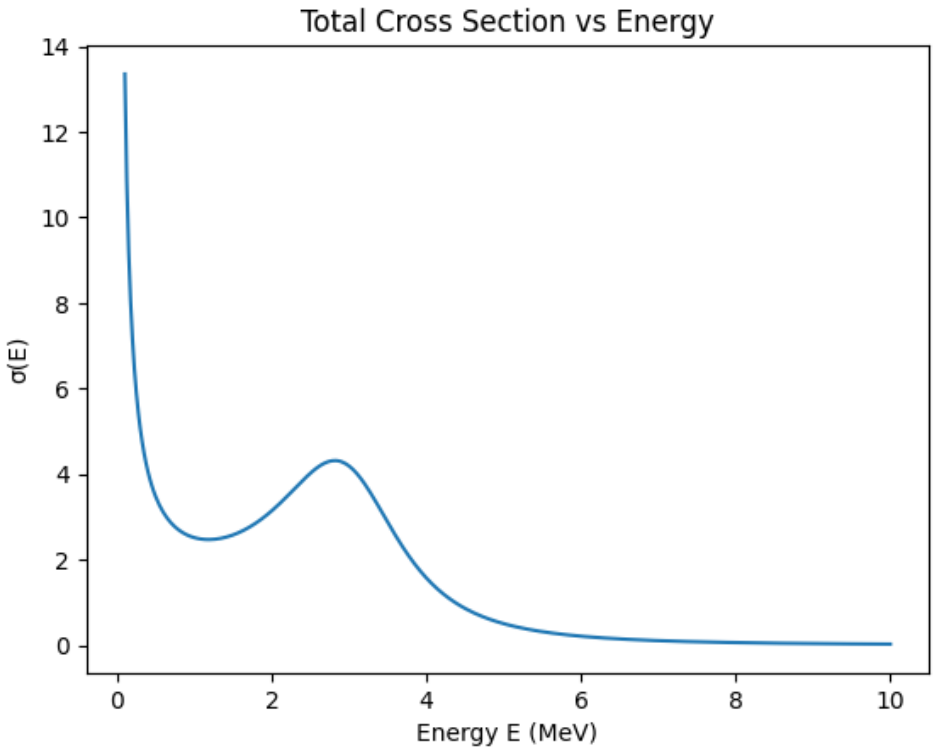


Figure 2: Total cross section as a function of energy E (in MeV).

Lecture 14–15
Final Review of Quantum Mechanics

1. General Structure of Quantum Mechanics

Quantum mechanics follows the scheme:

$$\text{Physical system} \;\Rightarrow\; \hat{H} \;\Rightarrow\; \hat{H}\psi = E\psi \;\Rightarrow\; \text{observables}$$

2. Fundamental Postulates

State:

$$\psi(\mathbf{r}, t)$$

Operators:

$$\hat{x} = x, \quad \hat{p} = -i\hbar\nabla, \quad \hat{H}$$

Expectation value:

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, dV$$

Time evolution:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

3. Operators and Commutators

$$[\hat{x}, \hat{p}] = i\hbar$$

4. Uncertainty Principle

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

5. Stationary States

$$\hat{H}\psi_n = E_n \psi_n$$

$$\psi(\mathbf{r}, t) = \psi_n(\mathbf{r}) e^{-iE_n t/\hbar}$$

6. Expansion in Eigenfunctions

$$\psi = \sum_n c_n \psi_n$$

7. Probability Current and Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

$$\rho = |\psi|^2$$

8. Typical Quantum Problems

- Potential step: reflection and transmission
- Tunneling: $E < U_0$
- Harmonic oscillator: discrete spectrum
- Central potential: separation of variables

9. Angular Momentum

$$\hat{L}^2 = \hbar^2 l(l+1), \quad \hat{L}_z = \hbar m$$

$$Y_l^m(\theta, \phi)$$

10. Hydrogen Atom

$$E_n = -\frac{13.6\,\text{eV}}{n^2}$$

Quantum numbers:

$$n, \quad l, \quad m$$

11. Spin and Total Angular Momentum

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

$$j = l \pm \frac{1}{2}$$

12. Approximation Methods

$$E_n = E_n^{(0)} + E_n^{(1)} + \dots$$

13. Scattering Theory

$$\psi \sim e^{ikz} + \frac{f(\theta)}{r} e^{ikr}$$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

14. Core Structure of the Course

1. Define Hamiltonian $\hat{H}$
2. Solve $\hat{H}\psi = E\psi$
3. Expand in eigenfunctions
4. Extract physical observables

15. Physical Interpretation

- Particle $\to$ probability wave
- Dynamics $\to$ evolution of amplitude
- Measurement $\to$ projection onto eigenstates

16. Connection with Other Fields

- Classical mechanics: limit $\hbar \to 0$
- Electrodynamics: potentials
- Statistical physics: distributions
- Nuclear physics: potentials and scattering

17. Final Statement

Quantum mechanics is:

$$\text{Operator theory} + \text{Spectral theory} + \text{Expansion in basis}$$